

UNIVERSITY OF THE PUNJAB

Fifth Semester – 2019

Examination: B.S. 4 Years Program

Roll No. in Fig.

Roll No. in Words.

PAPER: Sampling Techniques (Theory)
Course Code: STAT-305 Part-I (Compulsory)

MAX. TIME: 15 Min.

MAX. MARKS: 10

Signature of Supdt.:

Attempt this Paper on this Question Sheet only.

Please encircle the correct option. Division of marks is given in front of each question.

This Paper will be collected back after expiry of time limit mentioned above.

Q.1. Encircle the right answer, cutting and overwriting is not allowed. (1x10=10)

1. A complete list of all the units in the population is called -----
A. Sampling unit
B. Population unit
C. Sampling frame
D. None of the above
2. When the sample survey becomes a census survey, the sampling error becomes -----
A. Zero
B. One
C. Constant
D. None of the above
3. The Simple Random Sampling is a ----- technique.
A. Probability Sampling
B. Non-Probability Sampling
C. Both (A) & (B)
D. None of the above
4. Suppose we select every fourth invoice in a file. What type of sampling is this?
A. Simple Random
B. Cluster
C. Stratified
D. Systematic
5. Sampling in which sampling unit can be repeated more than once is called -----
A. Sampling with replacement
B. Sampling without replacement
C. Both (A) & (B)
D. None of the above
6. If a sample is drawn from each stratum minimizing the probabilities of non-preferred samples, it is known as:
A. Selection with proportional allocation
B. Controlled selection
C. Haphazard selection
D. None of the above
7. In systematic sampling, if N is not an integral multiple of nk , we use -----
A. Linear systematic sampling
B. Circular systematic sampling
C. Centrally Located systematic sampling
D. None of the above
8. For a population that is not normally distributed, the distribution of the sample means will -----
A. be negatively skewed
B. approach the normal distribution.
C. be positively skewed.
D. never approaches normal distribution.
9. The stratification after the selection of sample is called:
A. Two-way stratification
B. Deep stratification
C. Post stratification
D. None of the above
10. ----- is the standard deviation of a sampling distribution.
A. Sample standard deviation
B. Replication error
C. Meta error
D. Standard error



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PAPER: Sampling Techniques (Theory)

Course Code: STAT-305 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q. No. 2. Write a short note on the following:

(4x5=20)

- i. Determination of Sample Size
- ii. Two way Stratification
- iii. Linear Systematic Sampling
- iv. Proportional Allocation in Stratified Random Sampling

Q. No. 3. Show that the variance of the estimate of population total $\hat{y} = N\bar{y}$ from a simple random sample is $V(\hat{y}) = N^2(1-f)\frac{s^2}{n}$.

(07)

Q. No. 4. If the loss function due to an error in $\bar{y}$ is $\lambda|\bar{y} - \bar{Y}|$ and the cost function is $C = C_0 + C_1n$, then show that the most economical value of 'n' in simple random sampling, ignoring finite population correction is $(\frac{\lambda S}{C_1\sqrt{2\pi}})^{2/3}$

(07)

Q. No. 5. If the terms in $\frac{1}{N_h}$ are ignored relative to unity, show that for estimated mean from stratified random sample of size n_h ,

(09)

$$V_{opt} \leq V_{prop} \leq V_{ran}$$

where the optimum allocation is for fixed 'n'.

Q. No. 6. Show that the mean of a systematic sample is more precise than the mean of a simple random sample if and only if $S_{wsy}^2 > S^2$

(07)